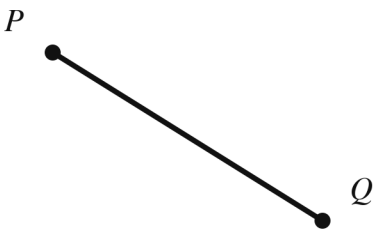
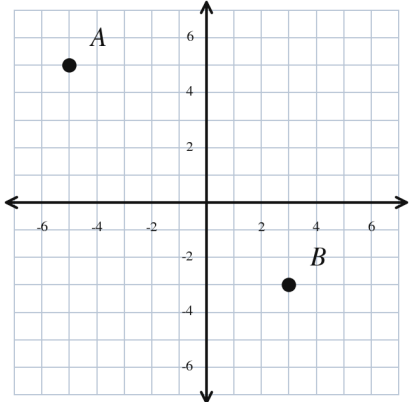


Part 1 – Construct the midpoint of each segment. Compass and straightedge only. Leave your arcs showing.

		
---	---	---

Part 2 – Find and label the midpoint. Use the midpoint formula, then plot your answer on the grid.

	a. Write the coordinates of each point. A = _____ B = _____	b. Use the midpoint formula to find M. $M = ((x_1+x_2)/2 , (y_1+y_2)/2)$ c. Plot and label M on the grid.
---	--	--

Part 3 – Find each distance. Use the distance formula. Leave irrational answers in simplest radical form AND as a decimal.

a. Find CD. C(-6, -3) D(6, 2) $d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$ = $\sqrt{(\text{rise}^2 + \text{run}^2)}$ <i>count the rise and run right off the grid</i>	b. Find EF. E(-2, 5) F(4, -1)	c. One of these came out a whole number and one did not. What has to be true about the rise and the run for the distance to be a whole number?
---	---	--

Part 4 – Team Challenge. Work with your group. Every member should be able to explain the answer.

	Three friends live at A, B, and C. They agree to meet at the midpoint of the two houses that are FARTHEST apart. a. Find all three distances.	b. Which two houses are farthest apart? _____ c. Find and label the meeting point. _____
--	--	--